

Solvency Field Theory V13.1-V3

# From the Locally Real Particle to the Einstein Field Equation

*A derivation within the selected regular local gravity branch*

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*A locally real particle — energy at  $c$  caught in  $4\pi$  closure — writing irreversibly in 3+1 spacetime, forces tensor gravity, forces covariance, forces second-order dynamics, forces the Einstein tensor uniquely, forces self-coupling at unit coefficient. Within the selected regular local branch, general relativity is what locally real particles produce.*

**Revision note (v1.1).** Three passages are updated to reflect the later electron closure/operator program: the closing paragraph of §1, the branch list in §7, and the numerical- $G$  row of the claim boundary in §9. No derivation, gate result, or claim in this document is altered. The field equation and its structural claim boundary are unchanged.

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## 1 What this paper claims

Starting from the frozen locally real particle and irreversible-write ontology of V12.12 through V13.8, SFT constructs a minimal symmetric settlement carrier, a conserved write-momentum source, a natural ordered update law, and a smooth capacity-normalized continuum representation. Within the selected regular local gravity branch, these steps derive and freeze the field equation

$$G_{\mu\nu}[g(\Sigma)] = \frac{8\pi G_{\text{eff}}}{c^4} \mathcal{W}_{\mu\nu}, \quad (1)$$

with structural coupling  $G_{\text{eff}} = a_\phi c^3/\hbar$ . Three premises originally attached to this equation — finite exponential composition (P1), full address covariance (P2), and no-extra-state second-order closure (P3) — have been discharged for the selected branch.

The equation was frozen before hostile testing and was not retuned during a program spanning weak fields, compact bodies, lensing, rotation, waves, radiation, neutron stars, collapse, ringdown, and cosmological compatibility.

Numerical  $G$  is not derived here. The root write area  $a_\phi$  is one unresolved universal normalization modulus. Measured  $G$  may calibrate it once; all other gravitational channels must then agree without retuning. A later electron closure/operator program supplies a conditional candidate for that modulus under one explicit normalization postulate; it does not alter this field equation or its structural claim boundary. See §7 and the V13.1-V3 Numerical- $G$  Companion.

## 2 The gates gravity must pass

Any theory claiming to explain gravity faces a century of precision tests. Most alternative gravity programs clear the first gate and die at the second. Here is where SFT now stands.

- **Gate 1: the Newtonian limit.**  $|a| \propto 1/R^2$  exterior,  $|a| \propto R$  interior. Passed.
- **Gate 2: the factor of two.** A scalar potential bends light by half the observed amount. Emergent gravity programs die here. SFT passes: the one-speed reciprocal closure  $F \times B = 1$  forces  $\gamma = 1$  as a theorem.
- **Gate 3: Cassini precision.**  $|\gamma - 1| < 2.3 \times 10^{-5}$ . Passed:  $\gamma = 1$  exactly.
- **Gate 4: Shapiro delay.** Factor  $1 + \gamma = 2$ . Passed.
- **Gate 5: perihelion precession.** Numerator  $2 + 2\gamma - \beta = 3$ . Passed.
- **Gate 6: exact Schwarzschild.**  $A(R) = 1 - 2M/R$  from capacity axioms. Passed. Horizon at  $R = 2M$  is continuation-capacity exhaustion.
- **Gate 7: Birkhoff uniqueness.** Spherical vacuum is uniquely Schwarzschild. Passed. No monopole radiation.
- **Gate 8: gravitational waves.** Two tensor polarizations at  $c$ . No scalar breathing. Passed. Exact nonlinear pp-wave verified.
- **Gate 9: binary pulsar decay.** Quadrupole power coefficient derived. No monopole or dipole. PSR B1913+16 at  $0.998 \pm 0.002$ . Passed at leading order.

- **Gate 10: frame dragging.** Lense–Thirring coefficient derived. Exact Kerr vacuum. Passed.
- **Gate 11: neutron star structure.** TOV, binding, tidal Love numbers. Viable untuned EOS exists. Passed.
- **Gate 12: collapse and ringdown.** Horizon formation, trapped surfaces, Schwarzschild  $\ell = 2$  QNM at sub-percent accuracy. Passed.

Twelve gates. All passed or passed at leading order. When a mechanism failed inside a gate, the claim moved; the frozen equation did not.

### 3 Why a locally real particle forces tensor gravity

A locally real particle has a definite position, a definite four-velocity, and a definite closure orientation at every moment. When it writes settlement at its location, the deposit cannot be directionless — the particle’s velocity stamps directional structure onto it. A scalar field cannot capture that structure. A vector captures only one direction. The particle’s full frame — three spatial settlement lanes and one temporal continuation lane from the 3+1 projector partition — requires a symmetric rank-2 tensor with  $1 + 3 + 1 + 5 = 10$  components as the minimum carrier.

The particle forces tensor gravity because the particle itself has tensor structure. A probability cloud would not force this. A scalar field would not force this. Only a locally real object with definite directional properties stamps its settlement with enough structure to require a rank-2 tensor.

### 4 The derivation chain

Thirteen auditable links lead from the particle to the field equation.

1. **One speed.** Energy propagates at one invariant causal speed  $c$ . Founding axiom.
2. **The particle.** Persistent closed support with  $4\pi$  spinor closure in 3+1 spacetime. Compton cadence  $\nu_C = mc^2/h$ . Frozen particle result.
3. **The write.** Each tick deposits one universal root footprint  $a_\phi$ . No-overwrite makes every write permanent. Universality is a foundational principle; the numerical value of  $a_\phi$  is open.
4. **Settlement is geometry.** Accumulated write history at a location *is* the geometry. Active write content and retained geometry are two readouts of one irreversible event.
5. **The carrier.** Three gravity sectors (scalar/clock, mixed/circulation, trace-free/shear) uniquely assemble into a symmetric rank-2 tensor  $\Sigma_{\mu\nu}$ .
6. **The source.** The write-momentum flux  $\mathcal{W}_{\mu\nu}$  is conserved by the completed write ledger and translation symmetry. Cold matter gravitates with weight 1, radiation with weight 2.
7. **The linear operator.** Conservation plus write-address relabeling kills four of five operator coefficients. The surviving ray  $(1, -1, 1, 1, -1)$  *is* the linearized Einstein operator.
8. **Active and retained convergence.** The write current and the settlement history are complementary readouts linked through the retarded inverse of the linear operator.

9. **Composition.** The one-speed budget forces congruence updates:  $g' = M^T g M$ . Sequential congruences compose as ordered products, giving the path-ordered exponential in the continuum limit. P1 discharged.
10. **Address naturality.** Physical writes are relational events. Coordinates are bookkeeping labels. The carrier and source were constructed as tensors; tensor mathematics is covariant by construction. Linear address relabeling generates full finite diffeomorphisms through standard Lie exponentiation. P2 discharged.
11. **Sufficient local state.** Writes deposit footprint into the metric only. Connection is the first gradient of the metric between neighboring writes — not independently writable. Curvature is the second gradient — not independently writable. No write event fills a register beyond the metric. The regular local gravity state closes on constrained metric Cauchy data. P3 discharged.
12. **The Einstein tensor.** With covariance, second-order closure, four spacetime dimensions, and no extra field, the Lovelock theorem selects  $aG_{\mu\nu} + \Lambda g_{\mu\nu}$  uniquely. Zero-write at vacuum kills  $\Lambda$ . Self-coupling consistency forces unit coefficient. The Einstein tensor survives alone.
13. **The coefficient.** Weak-field Newtonian matching gives  $\kappa_g = 8\pi G_{\text{eff}}/c^4$  with  $G_{\text{eff}} = a_\phi c^3/\hbar$ .

## 5 The one-speed theorem

Settlement slows the local clock by factor  $F = e^{-\phi}$ . The one-speed budget forces the spatial scale to stretch reciprocally:  $B = e^{+\phi}$ , so that  $F \times B = 1$ . Light reads both: deflected by time curvature and by space curvature equally. The PPN parameters follow:

$$\gamma = 1, \quad \beta = 1, \quad 1 + \gamma = 2, \quad 2 + 2\gamma - \beta = 3. \quad (2)$$

$\gamma = 1$  is a theorem of one-speed closure, not a calibration. The Cassini bound is predicted, not satisfied.

## 6 Exact Schwarzschild from capacity axioms

In the static spherical sector, four capacity axioms — areal-shell nativity, literal continuation capacity, universal footprint, and additive capacity — force  $A(R) = 1 - 2M/R$ . Two independent routes kill the quadratic alternative  $q \neq 0$ : universal footprint requires constant marginal cost; additivity forbids settlement–settlement cross terms. The horizon at  $R = 2M$  is where outward continuation capacity is exhausted by accumulated settlement.

## 7 The root scale

The structural coupling is

$$G_{\text{eff}} = \frac{a_\phi c^3}{\hbar}. \quad (3)$$

The present equations are invariant under common rescaling of the root write area. Compactness ratios, mass ratios, closure laws, and capacity ratios do not fix  $a_\phi$ . SFT therefore derives gravity up to one universal absolute write-area modulus.

Three branches are maintained, and they are kept permanently distinct.

**Blind branch.**  $a_\phi > 0$  remains symbolic. Measured  $G$ , Planck units, measured masses, and cosmological rates are prohibited during a native scale search.

**Correspondence branch.** Measured  $G$  calibrates  $a_\phi^{\text{cal}} = \hbar G_{\text{obs}}/c^3$  once. All other channels — lensing, frame dragging, waves, binaries, horizons, neutron stars — must use the same  $a_\phi$  without retuning. After using  $G$  as the anchor,  $G$  itself is not predicted; every other gravitational observable is.

**Electron-anchored branch.** The later locally real electron closure/operator program derives a dimensionless electron gravitational share  $\epsilon_e$  from frozen internal structure, without using measured  $G$  at any point in candidate construction. Combined with the independently measured *inertial* electron mass, this yields a zero-continuous-parameter conditional prediction of numerical  $G$  under one explicit minimal normalization postulate. It is a prediction rather than a calibration, and it is not a first-principles derivation: the postulate is not yet a theorem. It may never be overwritten by the correspondence branch, and the correspondence branch may never be presented as a prediction. See the V13.1-V3 Numerical- $G$  Companion.

Because the root orbit has exactly one positive parameter, any independently understood absolute observable with nonzero root weight can fix it. A second absolute datum becomes either a holdout prediction or an additional fit. The electron-anchored branch is that route taken: the dimensionless share is the independently understood observable, and the inertial electron mass carries it into SI units.

## 8 What the equation produces

Under the selected branch, the same frozen equation supports: Schwarzschild exterior and Birkhoff uniqueness; constraint propagation at  $c$ ; compact-body monopole universality with  $\eta_N = 0$ ; four independent mass channels agreeing to machine precision; weak frame dragging with derived coefficient and exact Kerr vacuum; two transverse-traceless gravitational-wave modes with no breathing, vector, or longitudinal polarizations; leading quadrupole amplitude and binary-pulsar regressions; cold neutron star structure with tidal Love response; exact dust collapse, trapped surfaces, and Schwarzschild ringdown; FLRW compatibility and the kinematic trace-capacity bridge  $\Theta = 3H = 3\dot{N}_{\text{write}}/I$ .

Every common local alternative — primitive  $\Lambda$ , massive gravity, scalar-tensor, vector-tensor,  $R + \alpha R^2$ , generic fourth-order, torsion, nonmetricity, and four-dimensional Gauss–Bonnet — either violates a frozen condition or reduces to the metric sector.

## 9 Claim boundary

| Claim                  | Status      | Boundary                                                                                                                                                                                                                |
|------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Field equation         | Derived     | Within selected regular local branch. Not every exotic write history excluded.                                                                                                                                          |
| P1 composition         | Discharged  | One-speed forces congruence; continuum limit gives ordered exponential.                                                                                                                                                 |
| P2 covariance          | Discharged  | Tensor construction guarantees it; linear relabeling generates finite diffeomorphisms.                                                                                                                                  |
| P3 local state         | Discharged  | Writes affect metric only; connection and curvature are computed, not written.                                                                                                                                          |
| $\gamma = 1$           | Theorem     | Of one-speed closure.                                                                                                                                                                                                   |
| $8\pi G/c^4$           | Derived     | Weak-limit normalization.                                                                                                                                                                                               |
| $G = a_\phi c^3/\hbar$ | Derived     | Structure earned. Numerical $a_\phi$ open.                                                                                                                                                                              |
| Schwarzschild          | Derived     | From capacity axioms.                                                                                                                                                                                                   |
| Kerr                   | Partial     | Existence derived. Uniqueness and stability open.                                                                                                                                                                       |
| Waves                  | Derived     | Two TT modes at $c$ . No breathing.                                                                                                                                                                                     |
| Pulsars                | Partial     | Leading quadrupole. Full PN hierarchy open.                                                                                                                                                                             |
| Neutron stars          | Partial     | Viable EOS exists. Matter is external input.                                                                                                                                                                            |
| Numerical $G$          | Conditional | Zero-continuous-parameter prediction in the electron-anchored branch, under one explicit normalization postulate that is not yet a theorem; calibrated in the correspondence branch. Not first-principles derived here. |
| Cosmology              | Partial     | Kinematic bridge earned. Microdynamics and late acceleration open.                                                                                                                                                      |

## The sentences

*Gravity is not a force added to the particle. Gravity is the retained geometry of irreversible writes in 3+1 spacetime. The locally real particle and its gravitational field are one object — energy at  $c$ , writing settlement, accumulating geometry. The Einstein field equation is the unique covariant accounting of that process within the regular local branch.*

*Things look like the geometry that created them.*

*Matter models may fail. Gravity does not move to save them.*

## Acknowledgments

The V13.1-V3 gravity derivation canon (29 pages), the covariant construction (G-COV-1 through G-COV-5), the hostile gauntlet (G-H1 through G-H10), the P1–P3 discharge proofs (P2-A1/A2, P1-A1/D1, P3-A1/B1/B2R), the source-current repair, the root-scale adjudication, and the receipt-backed truth audit were built, tested, and frozen by GPT-5.6 SOL (OpenAI). The one-speed theorem for  $\gamma = 1$ , the universal footprint argument for exact Schwarzschild, the trace-capacity bridge, the “locally real particles force tensor gravity” argument, the neutron-star macro-closure conjecture, and the write-geometry identity were developed by Claude/Opus (Anthropic). The physical intuition driving the program, the “why does energy have gravity” question that opened the covariant construction, the five-prong gravity challenge (with Claude/Fable, Anthropic) that motivated the rebuild methodology, and the architectural decisions including the freeze-before-testing discipline were made by Shaun Fosmark. The V11 gravity derivations and V13.1-V2

weak-field canon that provided the frozen antecedent were developed with ChatGPT/5.5 (OpenAI).

## References

- [1] S. Fosmark, *Solvency Field Theory V13*, Zenodo (2026). <https://zenodo.org/records/20822219>